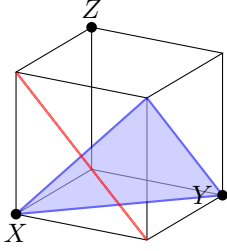


1 Exercise 1

consider the cube $C = (x, y, z) \in \mathbb{R}^3 | 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$ in $\mathbb{R}^3$. Find the angle between the plane P_1 containing the 3 points: $(0, 1, 0), (1, 0, 0), (1, 1, 1)$ and the plane P_2 containing the 3 points: $(0, 0, 0), (1, 1, 0), (1, 0, 1)$.



To find the angle between the two planes, we must firstly find the normal vector that the individual planes define and then find the angle between those two.

$$P_{1X} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad P_{1Y} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad P_{1Z} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Vectors for P_1

$$P_{1XY} = P_{1X} - P_{1Y} = \begin{pmatrix} 0-1 \\ 1-0 \\ 0-0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad P_{1XZ} = P_{1X} - P_{1Z} = \begin{pmatrix} 0-1 \\ 1-1 \\ 0-1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$$

The normal vector for plane $P1$ is the cross product of the two vectors in the plane.

$$\vec{N1} = P_{1XY} \times P_{1XZ} = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

The procedure goes for the second set of points, to generate the set of vectors, the plane they sit on, and the normal vector that extends from it.

Points

$$P_{2X} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad P_{2Y} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad P_{2Z} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

Vectors for $P2$

$$P_{2XY} = P_{2X} - P_{2Y} = \begin{pmatrix} 0-1 \\ 0-1 \\ 0-0 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}, \quad P_{2XZ} = P_{2X} - P_{2Z} = \begin{pmatrix} 0-1 \\ 0-0 \\ 0-1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$$

Normal vector from plane $P2$

$$\vec{N2} = P_{2XY} \times P_{2XZ} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$$

The angle between these two vectors will then indicate the angle between the planes $P1$ and $P2$

$$\theta = \cos^{-1}\left(\frac{N1 \cdot N2}{|N1||N2|}\right) \Rightarrow \frac{(-1)(1) + (-1)(-1) + (1)(-1)}{\sqrt{(-1)^2(-1)^2(1)^2} \cdot \sqrt{(1)^2(-1)^2(-1)^2}} \Rightarrow \frac{-1}{\sqrt{(3)^2}} \Rightarrow \cos^{-1}\left(\frac{-1}{3}\right) * \frac{180}{\pi} \approx 109.492$$

2 exercise 2

Find the center of mass $(\bar{x}, \bar{y}, \bar{z})$, for the wire C shaped as a helix, with density $p = kz$, (k is a constant). The wire C has parametrization:

$$x(t) = 3\cos(t), y(t) = 3\sin(t), z(t) = 4t, 0 \leq t \leq \pi$$

The following equations define the center of mass:

- The curve length $s(t)$, as a function of t satisfies: $\frac{ds}{dt} = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}$
- $m = \int_C p(x, y, z) ds$
- $\bar{x} = \frac{1}{m} \int_C xp(x, y, z) ds, \bar{y} = \frac{1}{m} \int_C yp(x, y, z) ds, \bar{z} = \frac{1}{m} \int_C zp(x, y, z) ds$

This is a rather convoluted question, essentially asking, if we have a piece of wire, in this case a helix, meaning there is a one large curve, since it is bounded to $0 \leq t \leq \pi$, what is the position in space of the center of its mass. The parametrization describes how the coil will expand if we increase t . The first two components x and y are trigonometric functions, causing the direction to be circular around the z -axis, and the z component simply indicates that it increases linearly with t .

Firstly we need to calculate the change in t into changes on the actual physical wire. We're given the density p and mass is the product of density and length. So the curve length is calculated as:

$$\frac{ds}{dt} = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}$$

differentiating the functions

$$x'(t) = \frac{d}{dt}x(t) \Rightarrow -3\sin(t)$$

$$y'(t) = \frac{d}{dt}y(t) = \frac{d}{dt}3\cos(t)$$

$$z'(t) = \frac{d}{dt}z(t) = \frac{d}{dt}4t$$

into the curve length

$$\frac{ds}{dt} = \sqrt{(-3 \cdot \sin(t))^2 + (3 \cdot \cos(t))^2 + (4)^2}$$

rearrange, use distributive property and recognize the idiot formula $\sin^2(t) + \cos^2(t) = 1$

$$= \sqrt{9 + 1 + 16} = \sqrt{25} = 5$$

$$ds = 5dt$$

Relationship for density is given, with z given and k is some constant, it can be rewritten as:

$$p = k \cdot z$$

$$p = 4tk$$

Substituted into the mass relationship, with a change of variable from the $\frac{ds}{dt}$ relationship, so that mass is now in terms of t :

$$m = \int_0^\pi (4tk)(5)dt \Rightarrow 10k\pi^2$$

Since the lower bound is zero and are a single term with factors, it gets reduced to zero. Each of the coordinates for the center of mass can now be evaluated.

$$\begin{aligned} \bar{x} &= \frac{1}{10k\pi^2} \int_0^\pi 3 \cdot \cos(t) \cdot 4t \cdot k \cdot 5dt \\ &= \frac{60k}{10k\pi^2} \int_0^\pi t \cdot \cos(t)dt \Rightarrow \frac{60k}{10k\pi^2} [t \cdot \sin(t) + \cos(t)]_0^\pi \end{aligned}$$

$$= \frac{6}{\pi^2} \cdot (-1) - \frac{6}{\pi^2} + \cos(0) = -\frac{12}{\pi^2}$$

The same can be done for the two other coordinates:

$$\begin{aligned}\bar{y} &= \frac{1}{10k\pi^2} \int_0^\pi 3 \cdot \sin(t) \cdot 4t \cdot k \cdot 5dt \\ &= \frac{60k}{10k\pi^2} \int_0^\pi t \cdot \sin(t) dt \Rightarrow \frac{6}{\pi^2} [\sin(t) - t \cdot \cos(t)]_0^\pi \Rightarrow \frac{6}{\pi} \\ \bar{z} &= \frac{1}{10k\pi^2} \int_0^\pi (4t) \cdot 4t \cdot k \cdot 5dt \\ &= \frac{80k}{10k\pi^2} \int_0^\pi t^2 dt \Rightarrow \frac{8}{\pi^2} [\frac{t^3}{3}]_0^\pi \Rightarrow \frac{8\pi}{3}\end{aligned}$$

Giving coordinates:

$$[\bar{x}, \bar{y}, \bar{z}] = [-\frac{12}{\pi^2}, \frac{6}{\pi}, \frac{8\pi}{3}]$$